



Karthik Nagavelly
Lead Cloud / DevOps Engineer
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Professional Summary:

Over 10+ Years of experience in the IT industry with specialized experience in **cloud & DevOps** like **Azure/AWS** and **GitHub/ GitLab** repositories, configuration management tools **Ansible** CI/CD tools **Jenkins/ ADO**, Infrastructure as Code **Terraform** and **Docker** to run and segregate containers, Orchestration tools like **Kubernetes** and **Azure DevOps**.

- I have expertise in working on **Microsoft Azure Services, Virtual Networks, Subnets, Virtual Machines, Resource Groups, Express Routes, Traffic Manager, VPN, Load Balancing, Application Gateways, Virtual Machine Scale Sets** and **Auto-Scaling**.
- Proficient in Azure development, adeptly utilizing Azure services such as Azure Web Applications, App Services, Azure Storage, Azure SQL Database, Virtual Machines, Fabric Controller, Azure AD, Azure Search, and Notification Hub. Extensive experience in leveraging Azure Service Fabric for packaging, deploying, and managing scalable and reliable microservices and containers.
- Hands-on experience designing Azure cloud models for establishing secure and cross-premise connectivity with Azure VPN gateway and Content Delivery Network.
- Proficient in a diverse range of Azure services including Compute (Web Roles, Worker Roles), Databases, Networks, Internet of Things (IoT), PowerShell Automation, Monitoring, Key Vault, and Scheduling."
- Experienced in designing, configuring, and deploying solutions on Microsoft Azure, utilizing ARM Templates, AZURE PowerShell Module, and Azure CLI with a specific focus on achieving high availability and implementing auto-scaling mechanisms."
- Proficient in establishing Azure DevOps CI/CD pipelines to facilitate the deployment of API Management services, configuring backend certificates, and managing backend configurations through the creation of ARM templates.
- Proficiently utilized Azure Kubernetes Service (AKS) to deploy managed Kubernetes clusters within Azure. Demonstrated expertise in creating AKS clusters through the Azure portal, employing versatile deployment methodologies including Azure Resource Manager (ARM) templates and terraform.
- Skilled in strategic planning, deployment, continuous monitoring, and maintenance of Azure cloud infrastructure, encompassing multiple VM nodes to meet environment-specific requirements.
- Proficiently utilizes Azure Monitor in tandem with Application Insights to detect and diagnose issues within applications and their dependencies. Skilled in correlating infrastructure-related problems by leveraging VM insights and Container insights.
- Hands-on experience in implementing Terraform modules to deploy diverse applications across multiple

cloud providers while effectively managing the associated infrastructure.

- Skilled in deploying Azure IaaS virtual machines (VMs) and cloud services (PaaS role instances) within secure VNets and subnets. Proficiently implemented Terraform modules to deploy applications across various cloud providers such as Azure and AWS.
- Proficient in deploying Azure IaaS virtual machines (VMs) and PaaS role instances within secure VNets and subnets. Demonstrated expertise in using Terraform modules to deploy applications across diverse cloud platforms, including Azure and AWS.
- Developed Kubernetes Cluster on Azure with master and Agent Node architecture and wrote YAML files

to create services like pods, deployments, auto-scaling, load balancers, labels, health checks, and Namespaces.

- Experienced in developing APIs using Kubernetes to manage and specify container instances for cloud-based server deployments. Proficient in scheduling, deploying, and managing container replicas within Kubernetes node clusters.
- Experienced in utilizing Git, GitHub, and Subversion (SVN) for branching, tagging, and maintaining version control consistency across diverse environments.
- Skilled in infrastructure development using a range of AWS Cloud Services such as VPC, Subnet, EC2, IAM, S3, RDS, ELB, Auto Scaling, CloudFront, CloudFormation, Elasticache, and AWS Import/Export
- Proficient in crafting AWS CloudFormation templates using both JSON and YAML formats, utilizing them as blueprints for building and deploying various AWS resources. Extensive experience includes working on Kinesis Data Streams and Kinesis Firehose, integrating them seamlessly with AWS Lambda for efficient serverless data collection.
- Versatile experience encompassing multiple facets of Jenkins, ranging from plugin management, securing, and optimizing performance to analytics, scaling, and integrating code analysis and testing phases for comprehensive CD pipeline execution within Jenkins.
- Proficient in creating Ansible Playbooks using YAML scripting and Puppet Manifests using DSL to provision a range of servers including Nginx, Apache Spark, Apache Web servers, and Tomcat servers.
- Extensive experience working with Windows Server 2016/2019 and conducting seamless SMS migration of Windows OS-based servers from VMware to the AWS console.
- Skilled in setting up, configuring, and troubleshooting TCP/IP, DHCP, DNS, NFS, NTP, and file-sharing servers such as CIFS and Samba on Linux platforms.
- Experienced in provisioning IaaS and PaaS resources, including Virtual Machines, Web, and Worker roles, utilizing both Microsoft Azure Classic and Azure Resource Manager (ARM). Proficient in deploying web applications on Azure through PowerShell Workflow
- Proficient in constructing deployment scripts and automated solutions using Shell scripting, PowerShell, and Python

Technical Skills:

Title	Tools Used
Cloud Environments	Microsoft Azure, AWS
Version Control Tools	GIT, Git Lab, Azure Repos, Bitbuckets
Configuration Management Tools	Ansible, Chef, Terraform, CloudFormation
Scripting/Programming Languages	Bash and Shell scripting, Python, Java
CI/CD Tools	GIT, BITBUCKET, Maven, ANT, Gradle, Jenkins, Azure Pipelines, GitLab Pipelines, SonarQube.
Build Tools	Maven, ANT
Container Tools	Docker, Kubernetes, ECS
Monitoring Tools	Splunk, Gary log, App Insights, ELK, Grafana, CloudWatch, Nagios
Operating Systems	Windows, UNIX, RHEL, CentOS& Ubuntu
Databases	MYSQL, NoSQL, MongoDB, Dynamo DB, MSSQL, PostgreSQL, Oracle
Web/Application Servers	Apache Tomcat, WebLogic, Oracle Application Server, Nginx, JBOSS

Professional experience:

Fidelity Investments (Dallas, Texas)

June 2023 to Present.

Senior DevOps Engineer

Responsibilities:

- Configured and deployed Azure Automation Scripts for an application, leveraging the Azure stack encompassing Compute, Web and Mobile services, Blobs, Resource Groups, Azure Data Lake, HDInsight Clusters, Azure Data Factory, Azure SQL, Cloud Services, and ARM services. Emphasized automation utilities and services.
- Configured and deployed Azure Automation Scripts for an application, leveraging the Azure stack encompassing Compute, Web and Mobile services, Blobs, Resource Groups, Azure Data Lake, HDInsight Clusters, Azure Data Factory, Azure SQL, Cloud Services, and ARM services. Emphasized automation utilities and services.
- Established and maintained CI/CD pipelines using Jenkins, automating the deployment of Confluent Cloud configurations and applications.
- Developed and configured HTTP Triggers within Azure Functions, integrated with Application Insights for comprehensive monitoring and load testing of applications using VSTS. Employed Python API to efficiently upload agent logs into Azure blob storage.
- Hands-on Experience in designing and implementing Service Oriented Architecture underlined with Ingress and Egress using Azure Data Lake Store & and Azure Data Factory by adding blobs to lakes for analytic results and pulling data from Azure Data Lake to the Blobs

analytic results and pulling data from Azure Data Lake to the Databricks.

- Used Azure blob to access required files and Azure Storage Queues to communicate between related processes. Configuring the Azure load balancer to load balance incoming traffic to virtual machines in Azure Cloud.
- Developed Infrastructure as Code (IaC) templates using Terraform to provision and manage cloud resources, including Kafka clusters, for consistent and reproducible deployments.
- Created Azure SQL database and performed monitoring and restoring of Azure SQL database. Performed migration of On-premises Microsoft SQL server to Azure SQL database.
- Deployed two-tier .NET, Python applications to Azure DevOps to focus on development by using services such as Repos to commit codes, Build Pipelines using PowerShell scripts for CI/CD and Test plans for unit test, deploy to App Service, Azure Application Insights collects health performance and usage data of the process, stored artifacts in blob storages.
- Created CI/CD pipelines for .NET, and python applications in Azure DevOps by integrating source codes GitHub, VSTS, and artifacts. Created deployment areas such as testing, pre-production, and production environment in Kubernetes cluster and configured App Insights, Prometheus, and Grafana monitoring dashboard for regular analysis.
- Configured and maintained CI/CD pipelines using Harness, automating the deployment of applications, and orchestrating testing processes for increased efficiency.
- Defined resources and specified deployment parameters to Azure Resource Manager (ARM) templates to create a resource group of virtual machines.
- Worked on Terraform key features such as Infrastructure as code, Execution plans, Resource Graphs, Change Automation, and Implemented Terraform modules for deployment of applications across Azure.
- Implemented Terraform modules for deployment of various applications in Azure cloud providers and managing infrastructure.
- Deployed microservices-based applications on Azure Kubernetes by using Azure Kubernetes Service (AKS), Ingress API Gateway, MySQL, SQL Databases, and Cosmos DB for stateless storage of external data, and set up reverse proxy Nginx servers and encrypting with SSL and TLS.
- Led the implementation of container-based applications on Azure Kubernetes using Azure Kubernetes Service (AKS) and Kubernetes Cluster for efficient cluster management. Configured Virtual Network for agent node deployment, integrated Ingress API Gateway, utilized MySQL Databases and Cosmos DB for

stateless external data storage and established reverse proxy Nginx within the cluster

- Implemented Jenkins pipelines into Azure pipelines to drive all microservices builds out to the Docker registry and then deployed to Kubernetes, Created Pods, and managed using Azure Kubernetes service (AKS).
- Demonstrated strong troubleshooting skills, resolving Kafka-related incidents promptly and conducting

root cause analysis for continuous improvement.

- Automated various infrastructure activities like continuous deployment, application server setup, and stack monitoring using Ansible playbooks.
- Building/Maintaining Docker container clusters managed by Kubernetes Linux, Bash, GIT, and Docker, on GCP.
- Used Chef for configuration management of hosted Instances within GCP. Configuring and Networking of Virtual Private Cloud (VPC).
- Ensured high availability for both IaaS VMs and PaaS role instances, enabling seamless access within the Virtual Network through the Azure Internal Load Balancer. Collaborated with business process managers, providing expertise in utilizing cutting-edge big data technologies such as Hive, Spark, Scoop, and NIFI for data ingestion and transformation, facilitating the creation of insightful business intelligence reports from vast data sets. Proficient in Python and Bash scripting, along with Apache Airflow for efficient job scheduling within GCP/Google's cloud-based environments
- Used Apache airflow in GCP composer environment to build data pipelines and used various airflow operators like bash operator, Hadoop operators, and python callable and branching operators.
- Worked on TERRAFORM for the provisioning of Environments in the GCP platform.
- Handle the various platforms like Linux, Windows, and GCP for automation purposes at the same time.
- Integrated Jenkins/Helm/Kubernetes/Vault with GCP to perform semi-automated and automated releases to lower and production environments.
- Worked closely with development teams to understand application requirements, ensuring seamless integration of Harness into the development and deployment lifecycle.
- Created Service accounts, VM instances, and VPC networks using Terraform on GCP.
- Automated Compliance Policy Framework for multiple projects in GCP.
- Deployed GKE on GCP with the help of Gitlab-Jenkins-Terraform integration.
- Conducted capacity planning exercises to ensure Confluent Cloud environments could handle anticipated workloads.
- Experience in Designing and implementing scalable cloud-based web applications using AWS and GCP.
- Migrated applications to the PKS and GCP cloud.
- **Environment:** Azure Cloud, Azure-DevOps, GCP, Terraform, AKS, Ansible, Maven, Azure-Monitoring, Power-Shell.

Responsibilities:

- Implemented AWS solutions using EC2, S3, RDS, DynamoDB, Route53, EBS, Elastic Load Balancer, and Auto scaling groups, which was done by creating AWS launch settings based on customized AMI. AWS IAM and security groups have been configured in VPC's public and private subnets.
- Worked on creating AWS Lambda functions to insert and retrieve data into DynamoDB. Worked with the architect to discuss the design and implementation to make sure data is stored from AWS S3 and DynamoDB securely.
- Manage AWS EC2 instances utilizing Auto Scaling, Elastic Load Balancing, and Glacier for our SIT, UAT, Prod, and DR environments as well as infrastructure servers for GIT and Puppet.
- Worked AWS Lambda to run servers without managing them and to trigger to run code by S3 and SNS and converting existing AWS infrastructure to serverless architecture deployed via gulp and AWS cloud formation.
- Created AWS Lambda Deployment function and configured it to receive events from AWS S3 buckets. Created Lambda functions and assigned roles in AWS Lambda to run Python scripts.
- Provisioned the highly available EC2 Instances using Terraform and cloud formation and wrote new plugins to support new functionality in Terraform.
- Involved in using Terraform to migrate legacy and monolithic systems to Amazon Web Services.
- Managed Kubernetes charts using Helm. Created reproducible builds of the Kubernetes applications, templated Kubernetes manifests, provided a set of configuration parameters to customize the deployment, and Managed releases of Helm packages.
- Implemented a production-ready, load-balanced, highly available, fault-tolerant, auto-scaling Kubernetes cloud infrastructure and microservice container orchestration.
- Created Clusters using Kubernetes and worked on creating pods, replication controllers, replica sets, services, deployments, labels, health checks, and ingress by writing Yaml files.
- Extensive experience in using Build Automation tools like ANT, Maven, Artifactory, and Jenkins
- Extensively worked on Jenkins continuous integration and for End-to-End automation for all builds.
- Worked on the end-to-end setup of the Artifactory pro as a container with a secure private Docker registry and local Docker repositories for storing the built Docker images.
- Worked on deployment automation of all the microservices to pull images from the private Docker registry and deploy them to the swarm cluster.
- Identified and implemented performance optimizations for Confluent Cloud, resulting in a Z% improvement in Kafka cluster throughput.
- Designed an ELK (Elastic Logstash Kibana) system to monitor and search enterprise alerts and

- Designed an ELK (Elastic Logstash Kibana) system to monitor and search enterprise alerts and configured ELK stack in conjunction with AWS and using Logstash to output data to AWS S3. Terminated SSL onto Nginx to remove overhead on API servers.
- Experience in creating Splunk and Data Dog dashboards and Alert's creation.
- Experience in complete software development life cycle (SDLC) with software development models like Jira, Agile Scrum Model, Prototype model & RUP, and Waterfall model.
- Installation and configuration of Apache Webserver and making junction work for the transportation of applications through a Webserver.
- Implementing a Continuous Delivery framework using Jira, SVN, Bamboo, maven & Nexus, and Puppet in a Linux environment.
- Hands-on experience in Diagnosing, and troubleshooting various networking, hardware & Linux / UNIX server services issues and performing preventive maintenance.

- Performing two types of Server migration one from Physical to Virtual (P2V) and another from Virtual to Physical (V2P).
- Perform software installations and upgrades to Linux systems and maintain them by established policies, procedures, and service-level agreements. System.
- Experience in deploying infrastructure as a code in AWS using Terraform, CloudFormation, and AWS SDKs like Python, node.js & and PowerShell.
- Worked with Ansible integrated with Terraform to deploy applications after infrastructure build using playbooks, creating roles for each application, error handling, and using Ansible-Lint.
- Worked on Ansible, Chef, terraform, Cloud Formation Templates.
- Experience in building the infrastructure by writing various Terraform/Cloud Formation templates.
- Configured Azure Infrastructure Automation using Runbooks and Terraform scripts.
- Involved in using Terraform to migrate legacy and monolithic systems to Amazon Web Services.
- Setting up the build and deployment automation for Terraform scripts using Jenkins.
- Provisioned the highly available EC2 Instances using Terraform and Ansible Playbooks.
- Developed environments of different applications on AWS by provisioning on EC2 instances using Docker, Bash, and Terraform.
- Demonstrated strong troubleshooting skills, resolving Kafka-related incidents promptly and conducting root cause analysis for continuous improvement.
- Managed AWS infrastructure as code using Terraform.
- Extensively involved in infrastructure such as code, execution plans, resource graphs, and change automation using Terraform.

automation using Terraform.

Environment: AWS Lambda, AWS S3, EC2, RDS, AWS Route53, Jenkins, Terraform, EBS, Linux, Ant, Maven, Data Dog, Puppet, Docker, Python, Ansible, Unix, Jira, Maven, GIT, Kubernetes, DynamoDB, Nginx, Splunk, ELK.

JDA Software Pvt Ltd
2021

November 2017 to August

DevOps Engineer

Responsibilities:

- Managed Azure infrastructure such as Azure Web Roles, Worker Roles, SQL Azure, Azure Storage, and Azure AD Licenses for development teams.
- Identity & Access Management: Azure Active Directory, Azure Identity, Multi-Factor Authentication (MFA).
- Implemented Cloud Solutions using various AWS Services including EC2, VPC, S3, Glacier, Security-Groups, CloudFront.
- Build Jenkins jobs and create Infrastructure from GitHub repositories containing Terraform code.
- Provisioned AWS networking Infrastructure such as VPC, subnets, and security groups using Terraform reusable modules utilizing Jenkins shared libraries.
- Virtualized the servers using Docker for the test environments and dev-environments needs as well as configured automation using Docker containers.
- Written Ansible Playbooks with Python SSH as the Wrapper to Manage Configurations of Azure Nodes and Test Playbooks on Azure instances using Python SDK.
- Developed tools using Shell scripting to automate some of the tasks. PowerShell scripting experience in creating automation processes for Branching, build creation, and deployments.
- Performed WebLogic Server administration tasks such as installing, configuring, monitoring, and performance tuning on a Linux environment.
- Extensive Experience working on Kubernetes to orchestrate the deployment scaling and management of docker containers.
- Experience in creating Docker images using Docker files, backing up using snapshots and managing the

images and Docker volumes, and writing JSON format scripts to automate and integrate Docker with Kubernetes.

- Proficient in various virtualization technologies, including VMware for traditional virtualization, Docker for containerization, and Kubernetes for container orchestration
- Used Jenkins pipelines to drive all microservices builds out to the Docker registry and then deployed to Kubernetes, Created Pods, and managed using Kubernetes.
- Building/Maintaining Docker container clusters managed by Kubernetes Linux, Bash, GIT, and Docker, on GCP.
- Used Docker to containerize Microservices in the cloud and managed clusters using Docker Swarm and Kubernetes. Developed Private cloud system solution on CoreOS using Kubernetes (Docker weave).
- Changing the AWS infrastructure Elastic Beanstalk to Docker with Kubernetes.
- Used Jenkins pipelines to drive all microservices builds out to the Docker registry and then deployed to Kubernetes, Created Pods, and managed using Kubernetes.
- Experience in Virtualization technologies like VMWare, and Vagrant and worked with containerizing applications like Docker, and Kubernetes.
- Leverage Docker containers and Docker consoles for managing the application lifecycle and worked on setting up the automated build on Docker HUB and deployed CoreOS Kubernetes Clusters to manage Docker containers with lightweight Docker Images as base files

Docker containers with lightweight Docker images as base files.

- Used Kubernetes to manage containerized applications using its nodes, Config Maps, selector, Services, and deployed application containers as Pods.
- Created and managed pods within Kubernetes, specializing in building production-grade kube clusters using KOPS (Kubernetes Operations)
- Managed local deployments in Kubernetes, creating local clusters and deploying application containers.
- Container management using Docker by writing Docker files setting up the automated build on Docker HUB and installing and configuring Kubernetes.
- Worked with Kubernetes to manage containerized applications using its nodes, ConfigMaps, selector, and Services and deployed application containers as Pods.

Environment: Azure cloud, Amazon AWS cloud, Bit-Bucket, Jenkins, Gradle, Kubernetes, Docker, Terraform, Ansible, Dynatrace, Shell Scripting, Linux.

Mercedes-Benz India Pvt. Ltd

May 2014-October 2017

Build & Release Engineer

Responsibilities:

- Configured SAML authentication to log in to our AWS accounts using single sign-on and deployed cross-account roles using AWS CloudFormation stack sets across different accounts from the master account.
- Automated cloud deployments using Python (boto & fabric) and AWS Cloud Formation Templates.
- Responsible for managing IAM Policies, providing access to different AWS resources, and designing and refining the workflows for grant access.
- Deployed and configured Elasticsearch, Logstash, and Kibana (ELK) for log analytics and application monitoring in integration with AWS Lambda and CloudWatch. Then store those logs and metrics into an S3 bucket using the Lambda function.
- Used Azure DevOps, GitHub Actions, and Octopus to build CI/CD pipelines.
- Implemented CI/CD for application using Jenkins as a CI tool for integrating different tools like GitHub, Maven, JUnit, and Nexus Artifactory and used Ansible as a configuration management tool for continuous deployment into testing, staging, and Production Environments.
- Create develop and test environments of different applications by provisioning Kubernetes clusters on AWS using Docker, Ansible, and Terraform
- Created Docker images using a Docker file, worked on Docker container snapshots, removed images, and managed Docker volumes.
- Installed Docker Registry for local upload and download of Docker images from Docker Hub and created Docker files to automate the process of capturing and using the images.
- Create, develop, and test environments of different applications by provisioning Kubernetes clusters on AWS using Docker, Ansible, and Terraform.
- Configured the Kubernetes provider with Terraform which is used to interact with resources supported by Kubernetes to create several services such as Deployments, services, ingress rules, Config Map, secrets, etc., in different Namespaces.

- Demonstrated strong troubleshooting skills, resolving Kafka-related incidents promptly and conducting root cause analysis for continuous improvement.

- Building/Maintaining Docker container clusters managed by Kubernetes. Linux, Bash, GIT, and Docker

- Building/Maintaining Docker container clusters managed by Kubernetes, Linux, Bash, GFI, and Docker. Utilized Kubernetes and Docker for the runtime environment of the CI/CD system to build, test deploy.
- Integrated Kubernetes with Hashicorp Vault to inject configurations at runtime for each service using init, config sidecars, and persistent volume sharing between app and config containers.
- To deploy instances using Ansible playbooks and write modules in Ansible to integrate with Apache Tomcat and AWS. And used the Ansible playbook to deploy applications.
- Designed an ELK (Elastic Logstash Kibana) system to monitor and search enterprise alerts and configured ELK stack in conjunction with AWS and using Logstash to output data to AWS S3. Terminated SSL onto Nginx to remove overhead on API servers.
- Worked with RedHat OpenShift Container Platform for Docker and Kubernetes. Used Kubernetes to deploy scale, load balance, scale and manage Docker containers with multiple namespace versions.

Environment: AWS, Azure, GitLab, Jenkins, Nexus, Go lang, GitHub Actions, Maven, Gradle, Docker, Kubernetes, Ansible, VMware, Web Logic, Tomcat, Cloud Watch, Perl, Oracle 10g/11g, GitHub, Grafana, Hashicorp Vault, Ruby, Git lab, Nagios, YAML, Splunk, JIRA, AppDynamics.

References: Available upon request

